Tejas Sriganesh

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Education

New York University,Tandon School of Engineering — **Aug 2025 – June 2027(Anticipated)**
Masters in Mechatronics, Robotics and Automation Engineering — *New York City, New York*

Birla Institute Of Technology and Science,BITS Pilani — **Aug 2021 – June 2025**
Bachelor of Engineering in Mechanical Engineering — *Pilani, India*

Research Interests

- **Research Interests** : Mobile Robotics, Swarm Robotics, Aerial Robotics, Robotic Manipulators, Humanoid Robotics, Motion Planning, Drone Manipulation, Deep Learning, Reinforcement Learning, Computer Vision

Professional Experience

Technische Universitat München-Munich Institute of Robotics and Machine Intelligence July 2024– December 2024
Robotics Engineering Intern, Munich, Germany | *Reinforcement Learning, 3D cadding, Electrical architechture design, Mobile Robotics*

* Worked on design, fabrication, inverse kinematics and motion planning of Quadruped Robot.
* Used Fusion 360 and 3D printing for design and assembly of a Quadruped Robot, design was inspired by Spot the quadruped.
* Created a portable power system for a quadruped and a Hexapod, that worked with a dc power source to power all robot systems
* Conducted research on robotic locomotion by applying Soft Actor-Critic (SAC) reinforcement learning via RL Baselines3 Zoo library to generate bezier gait patterns for the robots in multiple paths under Dr. Hoan on sim and real environments
- performed sim2real for robot between the pybullet environment and the hexapod itself by mapping joint angle values of sim to the dynamixel motors on the hexapod
* Setup indoor GPS system on the robot to be able to track location velocity and orientation of the bot in an indoor space

Atomic Energy Regulatory Board May 2022 – August 2022
Software Engineering Intern, Mumbai, India | *Python, Data Management, Sensor Control*

* Developed a predictive model for thermodynamic properties of D2 in a reactor environment based on coolprop python library.
* Model used inputs of pressure and enthalpy to get predictions of state variables such as temperature, entropy, pressure, volume, etc.

Research Projects

Gesture-Controlled humanoid finger | *Gesture Recognition, CV, Embedded Systems, Motor Control*

* Developed a real-time finger tracking system using MediaPipe to convert index-finger motion into motor control commands.
* Implemented real-time trajectory processing to convert finger motion into continuous control signals for motor actuation.
* Designed and built the complete electrical control system, including DC motor power delivery, H-bridge direction control, PWM-based speed regulation, and safe grounding between logic and power domains using an L298N driver and microcontroller
- Succesfully controlled finger using camera and mediapipe to mimic index finger actions with minimal lag

Multi-Robot Task Allocation and Path Planning | *Python, OpenCV, Software Engineering, linux, GIT*
Mentor : Dr. Avinash Gautam, Associate Professor, BITS Pilani

* Developed image segmentation algorithms for detailed environment mapping through color segmentation that detects a modal rgb value for each subsection to diffentiate freespace from obstacles and goalpoints over the map to convert an image into usable data.
* Optimized dynamic task transfer with A* path planning between any 2 goal points on the map. these were stored in a binary file for easy data recall that helped decrease path planning time and reduce penalties for overshooting deadlines
* Engineered a task transfer algorithm accounting for task compatibility and robot capacities
* Achieved a significant reduction in total penalties through strategic task assignments

Robotic Arm Pose Control | *Python, ROS, Arduino, Inverse Kinematics*
Mentor : Dr. M.S Dasgupta, Senior Professor, BITS Pilani

* Used Fusion 360 for 3-dimensional cad design and ANSYS for stress testing a robotic arm design over varied incremental loads
* Used ender 3 crealty to manufacture the prototypes to test out workspace for said robotic arm to make sure it could be able to reach all sections of the rover to perform complex tasks during competitions such as tightening bolts or being to pick up an object
* integrated arduino mega with a raspberry pi via serial connection using the pyserial library and ros 2 for smooth data processing
* created an inverse kinematics solver that took end effector values and plotted angles for joints to communicate to the actuators
* Successfully implemented inverse kinematics on robotic arm prototype to convert end effector positions to joint angles.

IoT-controlled Light Balance Act | *C++, Arduino, soldering, circuit design,IOT*

* Synchronized lighting effects with sound rhythm using timed relays for enhanced visual appeal.
* Built efficient circuits for powering and controlling lights, ensuring safety and reliability
* Programmed microcontrollers for seamless music-lighting synchronization through timed signals to relays.

Publications

Anubhav Elhance, Tejas Sriganesh, and Vinay Chamola. (2024). **Voronoi-based collaborative target tracking with drone swarms, edge prediction, and data fusion for enhanced efficiency**. *Communicated to IEEE Transactions on Vehicular Technology*, November 2024.

Leadership / Extracurricular

CRISS (Consortium for Research In Space Systems)

Mechanical Design Lead (June 2023 – Present)

* Collaborated on designing and manufacturing a robotic arm for a rover.
* Presented work at multiple competitions and secured notable global rankings in Rover Design Challenges (1st in IRDC 2023, 5st in ERC 2023 and 11th in IRC 2024)

Comedy & Performing Arts

Comedy Hub – Club President (Jan 2023 – July 2024) — *Mime Club – Writer/Actor (Jan 2022 – Present)*

* Led and organized stand-up comedy shows and open mics during campus cultural fest OASIS
* Performed stand-up comedy and sketch performances at cultural and technical festivals for live audiences of 300–2,000+ people